

CENTRALIA, IL, USA

WMO: 744657

Lat: 38.515N

Lon: 89.092W

Elev: 534

StdP: 14.41

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53887

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	12.1	-3.3	4.7	9.7	1.2	6.0	13.3	27.0	41.8	24.6	39.7	6.6	330	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	94.7	76.4	92.7	76.3	90.0	75.4	80.9	89.9	79.1	88.1	77.5	86.1	7.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	150.9	86.8	76.7	141.8	85.0	74.9	133.5	82.7	44.9	90.9	42.9	88.4	41.2	86.4	91.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.9	19.4	17.3	-0.3	98.6	4.8	3.0	-3.8	100.7	-6.6	102.5	-9.3	104.2	-12.8	106.4		
			-1.2	82.8	4.7	3.0	-4.7	85.0	-7.4	86.7	-10.1	88.4	-13.6	90.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.2	31.7	34.5	46.0	56.6	66.4	75.1	77.6	76.1	69.2	57.6	45.5	36.3
	DBStd	18.57	11.45	11.68	10.86	9.21	8.40	5.78	5.36	5.64	7.02	9.60	9.74	10.15
	HDD50	1940	575	450	208	39	1	0	0	0	0	32	198	437
	HDD65	4638	1032	854	592	276	90	4	0	3	36	274	588	889
	CDD50	4194	8	16	85	238	511	753	856	809	577	266	62	13
	CDD65	1418	0	0	4	25	134	308	392	347	163	43	2	0
	CDH74	13894	0	0	22	205	1108	2986	4170	3416	1585	387	15	0
	CDH80	5494	0	0	1	19	313	1194	1840	1446	578	102	1	0
Wind	WSAvg	6.9	9.1	8.6	8.9	8.7	6.9	5.8	4.6	4.0	4.3	6.3	7.7	8.0
Precipitation	PrecAvg	41.9	3.2	2.3	3.2	4.0	5.4	4.3	3.7	3.0	3.0	3.1	3.9	2.7
	PrecMax	56.2	7.6	4.6	6.9	10.0	14.2	9.8	8.5	6.6	8.0	6.2	7.2	8.3
	PrecMin	33.0	0.5	0.6	0.8	1.3	1.0	0.8	2.0	0.9	0.3	1.1	0.6	1.0
	PrecStd	5.9	1.9	1.2	1.5	2.3	3.3	2.2	1.8	1.4	1.9	1.5	1.9	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.1	69.3	77.9	82.4	90.5	95.4	100.2	98.9	93.7	88.2	75.2	65.8
		MCWB	60.3	61.8	63.6	67.3	73.7	74.0	74.7	76.8	72.7	72.4	63.0	60.5
	2%	DB	59.7	63.5	72.6	79.2	86.3	92.9	94.6	94.4	90.6	82.2	69.5	60.7
		MCWB	55.9	57.2	61.3	65.6	72.2	76.5	77.7	76.4	74.1	67.9	61.7	56.5
	5%	DB	54.8	57.9	67.9	76.5	83.6	90.5	92.6	91.2	86.5	78.4	64.8	56.4
		MCWB	50.5	51.9	59.0	64.2	70.9	74.8	77.3	75.7	73.0	66.2	57.8	51.8
	10%	DB	48.9	52.4	63.4	72.7	80.9	86.6	89.3	87.7	82.5	73.5	61.3	51.8
		MCWB	44.8	47.5	55.4	62.0	69.1	73.3	76.1	74.6	70.2	63.8	53.9	47.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.7	63.2	66.2	70.6	77.2	81.2	85.6	83.3	78.8	74.5	67.6	61.7
		MCDB	64.0	67.8	74.5	79.5	85.8	92.2	93.8	92.3	87.9	82.9	73.2	64.9
	2%	WB	56.8	58.8	62.7	68.0	74.7	78.9	81.4	80.4	76.9	71.8	62.7	57.0
		MCDB	60.0	62.6	70.6	77.1	83.1	89.5	89.5	89.5	86.0	79.6	67.1	60.4
	5%	WB	51.1	52.9	59.9	65.6	72.7	77.1	79.4	78.5	75.0	69.0	58.7	52.4
		MCDB	54.3	57.4	66.5	73.5	80.8	86.6	88.0	86.9	83.3	75.0	64.3	55.8
	10%	WB	44.0	47.4	56.8	63.2	70.6	75.0	77.6	76.6	72.7	65.5	54.7	47.4
		MCDB	48.7	52.7	62.4	71.3	78.4	83.2	85.9	84.5	79.7	72.2	60.5	50.4
Mean Daily Temperature Range	5% DB	MDBR	14.8	15.6	18.0	20.0	18.8	19.8	19.7	20.9	22.5	21.4	17.7	13.9
		MCDBR	20.6	22.7	22.2	23.6	21.3	22.8	22.7	24.3	25.6	25.1	21.9	18.9
		MCWBR	17.7	18.3	15.3	14.6	11.4	11.4	10.8	11.7	11.9	13.3	15.5	15.8
	5% WB	MCDBR	19.4	21.6	20.4	20.9	18.8	19.5	18.9	20.2	21.0	20.8	19.6	17.5
MCWBR		18.4	18.7	15.1	14.3	11.5	12.1	11.5	11.8	11.6	12.7	16.2	15.9	
Clear-Sky Solar Irradiance	taub	0.309	0.322	0.361	0.400	0.435	0.437	0.464	0.462	0.408	0.357	0.318	0.304	
	taud	2.456	2.405	2.361	2.254	2.213	2.233	2.159	2.154	2.294	2.436	2.489	2.482	
	Ebn at Noon	276	286	285	279	270	268	259	257	265	270	270	269	
	Edh at Noon	26	31	36	43	45	44	47	46	38	30	25	23	
All-Sky Solar Radiation	RadAvg	635	875	1185	1564	1755	2032	1976	1789	1494	1083	730	534	
	RadStd	65	95	113	90	154	132	130	100	101	113	80	54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140

Nomenclature: See separate page